T. Ben Baker, PhD, JD

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ACADEMIC APPOINTMENTS

- **Colby College, Assistant Professor** *2023–present*
  Waterville, ME
  – Department of Philosophy, Davis AI Institute
- **University of Pennsylvania, Provost Postdoctoral Fellow** *2020–2023*
  Philadelphia, PA
  – Körding lab, Departments of Bioengineering and Neuroscience

EDUCATION

- **University of Pennsylvania, PhD Philosophy** *2020*
  Philadelphia, PA
  – Dissertation: "Cognition in Nature: Information, Explanation, and Embodiment"
  – Social, Cognitive and Affective Neuroscience, Graduate Certificate
  – Dean's award for distinguished teaching by a graduate student
- **Yale Law School, JD** *2014*
  New Haven, CT
  – Substantial Analytic Writing on Mens Rea and Manslaughter
  – Lowenstein International Human Rights Clinic
- **Brown University, BA (Concentrations in Philosophy (honors) and Economics)** *2010*
  Providence, RI

PUBLICATIONS

Peer-Reviewed Articles & Book Chapters

- **Rethinking Artificial General Intelligence [link]** *2026*
  Invited book chapter, Introducing Philosophy of Mind, Today (eds. D Curry, L Daoust), Routledge
- **Use and Usability: Concepts of Representation in Philosophy, Neuroscience, Cognitive Science, and Computer Science [link]** *2026*
  with R Lange, A Richmond, R Cao, N Kriegeskorte, O Schwartz, X Pitkow, Neurons, Behavior, Data Analysis, and Theory
- **Understanding Agent Complexity Through Affordances [link]** *2025*
  with Sonia Roberts, Minds and Machines
- **Computational Kinematics of Dance: Distinguishing Hip Hop Genres [link]** *2024*
  with T Liu, J Matelsky, F Parodi, B Mensh, J Krakauer, K Körding, Frontiers in Robotics and AI
- **Three Aspects of Representation in Neuroscience [link]** *2022*
  with B Lansdell and K Körding, Trends in Cognitive Sciences
- **Creativity in AI-Dance Art [link]** *2022*
  Proceedings of the International Conference on Computational Creativity
- **Natural Information, Factivity, and Nomicity [link]** *2021*
  Biology & Philosophy
- **Referee Report of (hypothetical) Philosophy101 Textbook [link]** *2021*
  Teaching Philosophy
  – Spanish translation by Fredy H.P. Galindo, 2023, *Cuestiones de Filosofía*

Under Review & In Preparation

- **Situating AI Ethics in Language Education: Justice and Autonomy Across Learning and Assessment** *revised and resubmitted*
  with C Wang (first author), International Journal of Applied Linguistics
  – Argues that normative principles and situated ethical practice are complementary in AI ethics: fairness metrics for AI language assessment presuppose what the construct can register of a multilingual learner, datafied personalization reshapes learner autonomy, and institutions deploying such systems owe learners answerability

- **Six Dimensions of Dance Cognition** *in preparation*
  Philosophy of cognitive science article
  – An account of six distinct agent-level abilities engaged in dance — articulated primarily through Black social dance styles — offering dance scholarship a tool for analyzing the thinking in different practices, and cognitive science a detailed map of an activity often invoked to show what mind owes to body

- **Felt, Seen, and Measured** *in preparation*
  Self-assessment, observer perception, and high-speed kinematics of learning a dance phrase
  – Eleven dancers learned a hip-hop phrase from its choreographer and performed it under a four-camera high-speed markerless motion-capture rig; the paper maps where felt improvement, observer judgments, and measured kinematics (deviation, timing, amplitude, predictability) agree and diverge

Public Writing & Reports

- **Commentary on Favela and Machery [link]** *2023*
  Symposium contribution, The Brains Blog
- **The Time is Ripe to Reverse Engineer an Entire Nervous System [link]** *2023*
  Position paper, Alliance for C. elegans Neural Dynamics (ACeND)
- **What Does Engaged Philosophy Look Like? [link]** *2016*
  with L Daoust and R Willison, guest post, DailyNous.com
- **Before Their Time: Challenges to the Prohibition Against Child Marriage in Sierra Leone [link]** *2013*
  with A Elbogen, S Keene, Allard K. Lowenstein International Human Rights Clinic

TEACHING

Instructor of record for 17 semester-courses (300+ students through Spring 2026). Mean instructor rating 4.45/5 across evaluated sections. Frequency in parentheses where > 1.

Courses

- **Colby College, Assistant Professor, Department of Philosophy** *2023–present*
  Waterville, ME
  – All courses developed and first-taught by me
  – Central Philosophical Issues: Mind, Body & Machine ($\times 6$)
  – Philosophy of Dance ($\times 2$)
  – AI and Human Values
  – Philosophy of Emotion
  – Philosophy of Cognitive Science
  – Philosophy of Science
  – Bodies and Power
- **University of Pennsylvania, Instructor of Record, Department of Philosophy** *2018–2020*
  Philadelphia, PA
  – Introduction to Philosophy ($\times 2$)
  – Philosophy and Visual Perception
  – Ethics

Student Advising

- **Honors theses:** 1 in Philosophy; 1 in Computer Science (cross-departmental)
- **Senior independent studies:** 2 in Philosophy
- **Undergraduate research assistants:** 4 students, 2 in philosophy and 2 on the motion-capture pipeline and dance data analysis

Other Teaching

- **Davis AI Summer Bridges** (Summer 2026, *Colby*): Guest instructor for a full-day session with 30 students of mixed coding experience — analysis of 3D dance-motion data with interpretable movement features, a hip-hop dance session, and a technology-ethics discussion
- **Hip-hop dance instruction** (2011–present): workshops at Colby for SOBLU (Students Organized for Black and Latinx Unity; Fall 2025) and the Department of Performance, Theater, and Dance pre-semester dance intensive (Summer 2025); classes at Philadelphia studios including I Am Phresh Dance Academy and Urban Movement Arts (2015–2018); choreographer for student dance companies at Yale and Penn

TALKS

Invited Talks

- **Dancing How? Cognitive Dimensions and Kinematic Measures** *2027 (forthcoming)*
  Invited speaker, Creative Movement and Brain Health, ETH Zürich & UCL, Monte Verità, Ascona, Switzerland
- **Commentator, Symposium Session** *2027 (forthcoming)*
  American Philosophical Association, Eastern Division, Boston MA
  – Commentary on J Yi, "The Responsible Architect: An Existential-Hermeneutic Ethics for AI in Design"
- **Smaller Than Its Parts? Inter-level Dependence in Dynamical Systems** *2026 (forthcoming)*
  Paper jam (works-in-progress workshop), University of Edinburgh
- **Commentator, "AI, Stories and Theories"** *2026*
  American Philosophical Association, Eastern Division, Baltimore MD
  – Commentary on I Oved et al., "Virtual Embodied AI for a More Rigorous Philosophy and Science of the Mind"
- **What Does It Mean to Be Human?** *2025*
  Invited all-school speech, Loomis Chaffee School, Windsor CT
- **Rethinking Artificial General Intelligence** *2025*
  Oberlin Colloquium in Philosophy, Oberlin College
- **Invited Commentator, Symposium on Technology and Cognition** *2025*
  Southern Society for Philosophy and Psychology, annual meeting
- **Moving Between Genres: A Computational and Embodied Exploration of Dance** *2025*
  Invited contributor, Slippage: 3D Humanities Series, Northwestern University
- **Dance Machines!** *2024*
  James Madison University, sponsored by the Philosophy & Religion Department and JMU Dance Program
- **Six Forms of Dance Cognition** *2023*
  Mind the Body, University of Utah

Conference Presentations & Panels

- **Moderator, "Did Big Tech Capture the Government?"** *2026*
  with Katie Drummond (WIRED), In the News, Goldfarb Center for Public Affairs, Colby College
- **Ethics of AI-Mediated Language Education: Critical, Ecological, and Technomoral Perspectives** *2026*
  with C Wang, Artificial Intelligence Research in Applied Linguistics (AIRiAL) Conference, New York NY
- **So You Think AI Can Dance?** *2022*
  Eye, Mind, and Image: Themes from Visual Studies, UPenn
- **Panelist, Symposium on Affirming Black Excellence in Neuroscience** *2021*
  Cognitive Neuroscience Society (virtual)
- **Representation in Neuroscience** *2020*
  Neuromatch Conference 3.0 (virtual)
- **The Factivity and Nomicity of Information** *2019*
  Southern Society for Philosophy and Psychology, Cincinnati OH
- **Neural Dynamics and Inner Representations** *2017*
  Conference on Representing Reality, SUNY Potsdam
- **Learning to Dance with Dynamic Systems** *2017*
  Graduate BEN Talks, UPenn (internal)
- **Weakness of Will in Desire-First Accounts of Rational Action** *2015*
  Philosophers' Cocoon Conference, Tampa FL

COMPUTATIONAL KINEMATICS OF HUMAN MOVEMENT

- **Motion Capture & Data Collection (Kinelab)** *2024–present*
  Colby College
  – Designed and built a markerless motion-capture system at Colby; data collection ongoing in collaboration with Veronica Romero (Colby Psychology) under a joint umbrella IRB
- **Movement Modeling (KineTrace)** *2025–present*
  Colby College
  – Developing self-supervised next-window-prediction models trained on a custom biomechanical featurization of human movement (precursor: *Frontiers* 2024); early-stage collaboration with Amy LaViers (Robotics, Automation, and Dance Lab, Germantown PA)
- **Responsible AI Grant ($10,000)** *2026*
  Davis Institute for Artificial Intelligence, Colby College
  – Supporting sabbatical research on responsible computational kinematics — algorithmic bias, interpretability, and movement privacy in markerless motion capture
- **Technical Working Group, NEA Research Lab** *2026–present*
  Emory University (PI: Madeleine E. Hackney)
  – Advisory group for a National Endowment for the Arts Research Lab on arts-based rehabilitation for people with Parkinson's disease and dementia — intergenerational partnered dance, and gait, timing, and rhythmic-perception training; with Science Gallery Atlanta and Rockefeller University

MEDIA COVERAGE

- *The Chronicle of Higher Education*, LLM roleplay assignment featured (June 2026)
- *Dance Magazine*, feature on AI, motion capture, and dance [link]
- *Colby Magazine*, feature on the dance-genre classifier
- SciTube, educational video episode on *Computational Kinematics of Dance* (*Frontiers in Robotics and AI*, 2024)

PROFESSIONAL SERVICE & LEADERSHIP

- **Departmental/College Service, Colby College** *2023–present*
  – **Department of Philosophy** (2023–present): Clark Award and Pepper Prize judging (reading and scoring senior essays, coordinating prize text, presenting at the awards banquet); co-organizing the annual Philosophy Department Awards Banquet; reviewing honors-thesis proposals for the Honors Program
  – **Faculty Advisory Committee, Davis Institute for Artificial Intelligence** (2026–present): Planning around Colby's AI policies and tools, event organization
  – **Reuman Reading Group Organizer** (2024–present): communications, documentation, and publicity for Colby's 27-year-running faculty-student reading group in philosophy
  – **Search Committees** (2023–2026): three for hires in Philosophy (one TT, two VAP); one for the Director of the Davis AI Institute; one as external reviewer for Psychology
  – **Lunder Institute for American Art, Faculty Advisory Group** (2024–2025): Planning and participating in initiatives around integrating museum resources and visiting fellows into the work and teaching of Colby faculty, handling internal grant funding
  – **Science, Technology, and Society Faculty Advisory Board** (2023–2025): Contributing to initiatives for the STS department around student participation, course options, and internal funding
- **Conference/Event Organization** *2018–present*
  – **Philosophy Symposium** (Spring 2025, *Colby*): Organized annual public lecture on Ethics of Social AI, with student participation, sponsored by the Philosophy department and the Davis AI Institute
  – **The Flavor Continues Residency Organizer** (2024, *Colby*): Coordinated residency for a Boston-based Hip Hop non-profit, offering community workshops, guest teaching in Colby dance classes, and a performance
  – **Birge Lecture Co-organizer** (Fall 2023, *Colby*): Helped select and publicize the speaker for Colby's annual Birge Lecture
  – **Philosophy and Engagement Conference Co-organizer** (2016, *UPenn*): Interdisciplinary conference on philosophy and community engagement
  – **Indian Philosophy Reading Group** (2019, *UPenn*): Logistics and reading selection for a graduate discussion group

– **Admitted Student Weekend Coordinator** (2018, *UPenn*): Accommodations, activities, and faculty meetings for prospective students

- **Peer Review** *ongoing*
  – **Reviewer:** *Philosophical Psychology*; *Philosophy of Science*; *IEEE Robotics and Automation Magazine*; *Dance Research Journal*